

MEDICAL EQUIPMENT

Standards & Interoperability Study

HORIBA ABX Micros 60 · Abaxis Piccolo Xpress

Analysis of FHIR, HL7, LOINC Support & Generated Data

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1. Executive Summary

This study evaluates two widely used clinical laboratory analyzers — the HORIBA ABX Micros 60 hematology analyzer and the Abaxis Piccolo Xpress point-of-care chemistry analyzer — from the perspective of health data interoperability standards. Specifically, it examines their support for HL7 messaging, FHIR R4 compatibility, and LOINC coding of generated laboratory data.

Both instruments are designed for laboratory and point-of-care settings and generate structured clinical data that must be transmitted to Laboratory Information Systems (LIS) and Electronic Medical Record (EMR) systems. The degree to which they support modern health data standards directly impacts their integration with hospital IT infrastructure and their suitability as reference points for the Smartband V001 project.

Criterion	ABX Micros 60	Piccolo Xpress
Device Type	Hematology Analyzer (CBC)	Point-of-Care Chemistry Analyzer
Manufacturer	HORIBA ABX SAS (HORIBA Medical)	Abaxis Inc. (now Abbott Point of Care)
HL7 Support	Yes — via LiteDM software (HL7 bridge)	Yes — native HL7 output
FHIR R4 Support	Indirect — via middleware/integration	Indirect — via middleware/integration
LOINC Coding	Supported via LIS integration	Supported via LIS/EMR integration
LIS Connectivity	RS-232 / Ethernet via LiteDM	USB / Ethernet bidirectional
Data Format	ASTM, HL7 (via LiteDM)	ASCII, ASTM, HL7, XML
Direct FHIR API	Not natively supported	Not natively supported

2. HORIBA ABX Micros 60 — Hematology Analyzer

2.1 Product Overview

The HORIBA ABX Micros 60 is a compact benchtop hematology analyzer designed for small to medium clinical laboratories. It performs a Complete Blood Count (CBC) with 3-part leukocyte differential using electrical impedance technology and photometric hemoglobin measurement. It is part of HORIBA's ABX Micros Series, manufactured by HORIBA ABX SAS in France.

The analyzer processes up to 60 samples per hour in open tube mode and 55 samples per hour in closed tube mode. It requires only 10 µL of whole blood per cycle, making it suitable for pediatric and microsampling applications. The device uses stepper motor technology instead of a compressor, resulting in reduced noise and maintenance requirements.



2.2 Technical Specifications

Parameter	Value
Throughput	60 samples/hour (open tube) / 55 samples/hour (closed tube)
Sample Volume	10 µL whole blood (CBC and CBC 3-part DIFF)
Measurement Method	Impedance (WBC, RBC, PLT) + Photometry (HGB at 555 nm)
HCT Measurement	Numeric integration
3-Part DIFF Method	Impedance with specific lyse (LMG)
Parameters	18 parameters: WBC, RBC, HGB, HCT, MCV, MCH, MCHC, PLT, MPV, PCT, PDW, RDW, LYM%, LYM#, MON%, MON#, GRA%, GRA#
Memory	Last result + histograms (expandable with optional smart cards: 78 results)
Data Output	RS-232 (unidirectional); LiteDM software for LIS/EMR connectivity
Display	2-line LCD, 40 alphanumeric characters per line
Printer	Dot matrix printer
Power Supply	90–240 VAC, 50–60 Hz, 230 VA maximum
Dimensions	44 cm (H) × 36 cm (W) × 33 cm (D)
Weight	14 kg
Operating Temp	16–34°C, max 80% relative humidity
Certifications	UL 61010-1, CAN/CSA-C22.2 No. 61010-1

2.3 Generated Data — 18 CBC Parameters

The ABX Micros 60 generates the following 18 hematological parameters per blood sample:

Parameter	Abbreviation	LOINC Code	Unit	Clinical Significance
White Blood Cell Count	WBC	6690-2	10 ³ /μL	Infection, immune status, leukemia screening
Red Blood Cell Count	RBC	789-8	10 ⁶ /μL	Anemia, polycythemia screening
Hemoglobin	HGB	718-7	g/dL	Anemia, oxygen-carrying capacity
Hematocrit	HCT	4544-3	%	Anemia, dehydration assessment
Mean Corpuscular Volume	MCV	787-2	fL	Anemia classification (micro/macrocytic)
Mean Hemoglobin Corp.	MCH	785-6	pg	Anemia subtype classification
Mean Corp. Hgb Concentration	MCHC	786-4	g/dL	Hemoglobin concentration per cell
Red Cell Distribution Width	RDW	788-0	%	Anemia type differentiation
Platelet Count	PLT	777-3	10 ³ /μL	Clotting disorders, thrombocytopenia
Mean Platelet Volume	MPV	32623-1	fL	Platelet function, cardiovascular risk
Platelet Crit	PCT	51637-7	%	Platelet mass in blood
Platelet Distribution Width	PDW	32207-3	%	Platelet size variation
Lymphocyte %	LYM%	736-9	%	Viral infections, immune response
Lymphocyte #	LYM#	731-0	10 ³ /μL	Absolute lymphocyte count
Monocyte %	MON%	5905-5	%	Inflammatory conditions
Monocyte #	MON#	742-7	10 ³ /μL	Absolute monocyte count
Granulocyte %	GRA%	764-1	%	Bacterial infections
Granulocyte #	GRA#	30180-4	10 ³ /μL	Absolute granulocyte (neutrophil + eosinophil + basophil)

2.4 HL7 Support

The ABX Micros 60 itself does not natively output HL7 messages. HL7 connectivity is provided through the optional LiteDM (Lite Data Manager) software, which acts as a middleware bridge between the analyzer and LIS/EMR systems.

The LiteDM v3 Data Manager software enables HL7-based connectivity to EMR systems. Confirmed compatible EMR interfaces include Athena, Flatiron, Epic, eClinicalWorks, and CureMD. The connection is established via an HL7 bridge that translates the analyzer's RS-232 ASTM output into HL7 v2.x messages for transmission to the LIS.

Key HL7 capabilities via LiteDM:

- Bidirectional HL7 v2.x messaging between analyzer and LIS/EMR
- OBR segment carries the test order information with CBC panel codes
- OBX segments carry individual parameter results with LOINC codes as observation identifiers
- ADT messages for patient demographic integration
- QC data transmission to the LIS for quality management
- Confirmed interfaces: Athena, Flatiron, Epic, eClinicalWorks, CureMD

2.5 FHIR Support

The ABX Micros 60 does not natively support FHIR R4. There is no built-in FHIR API or FHIR resource generation in the device or LiteDM software. FHIR integration requires an external middleware layer or integration engine.

FHIR integration pathway for ABX Micros 60:

- Step 1 — ABX Micros 60 sends ASTM/HL7 v2.x data to LiteDM via RS-232
- Step 2 — LiteDM forwards HL7 v2.x messages to LIS/EMR via HL7 bridge
- Step 3 — An integration engine (e.g., Mirth Connect, Azure Health Data) transforms HL7 v2.x to FHIR R4 DiagnosticReport and Observation resources
- Step 4 — FHIR R4 resources are stored in a FHIR server (e.g., HAPI FHIR) for downstream consumption

When transformed to FHIR R4, each CBC result would generate a DiagnosticReport resource containing multiple Observation resources — one per parameter — each coded with the appropriate LOINC code.

2.6 LOINC Support

LOINC coding is not performed by the ABX Micros 60 analyzer itself. LOINC codes are assigned during LIS/EMR integration, either manually configured in the LiteDM interface mapping or automatically applied by the receiving LIS. The table in Section 2.3 lists the correct LOINC codes for all 18 parameters generated by the analyzer.

In HL7 v2.x messages, the OBX-3 field (Observation Identifier) carries the LOINC code in the format: LOINC_code^parameter_name^LN. For example, the WBC result would appear as: 6690-2^Leukocytes^LN.

2.7 Connectivity Architecture

Connection Type	Protocol	Direction	Notes
RS-232 Serial	ASTM E1394	Unidirectional (out)	Base hardware output — sends patient and QC results to LIS
RS-232 via LiteDM	HL7 v2.x	Bidirectional	Via LiteDM bridge — full LIS/EMR integration
Ethernet via LiteDM	HL7 v2.x	Bidirectional	LAN connectivity for multi-analyzer configurations
Smart Card (opt)	Proprietary	Local storage	QC and patient data storage — 78 results capacity

Connection Type	Protocol	Direction	Notes
LIS Integration	ASTM / HL7 v2.x	Bidirectional	Via LiteDM: Athena, Flatiron, Epic, eClinicalWorks, CureMD
FHIR R4	Not native	—	Requires external HL7→FHIR transformation middleware

3. Abaxis Piccolo Xpress — Point-of-Care Chemistry Analyzer

3.1 Product Overview

The Abaxis Piccolo Xpress is a fully automated, portable point-of-care blood chemistry analyzer designed for use in any patient-care setting. It provides lab-accurate clinical chemistry results within 12 minutes using only 100 µL (0.1 cc) of whole blood, serum, or plasma — equivalent to approximately 4 drops of blood.

The system uses patented single-use self-calibrating reagent discs, each containing all reagents required to perform a complete panel of up to 14 tests simultaneously. Abaxis was acquired by Zoetis in 2018; the human medicine product line (Piccolo Xpress) is now distributed in the USA by Abbott Point of Care. The device is widely used in primary care, urgent care, hospitals, military settings, and research facilities.



3.2 Technical Specifications

Parameter	Value
Test Time	12 minutes per complete panel
Sample Volume	100 µL (0.1 cc) — whole blood, serum, or plasma
Sample Type	Lithium heparin whole blood, serum, or plasma
Available Panels	17 pre-configured multi-analyte panels
Total Tests	31 unique chemistry tests across all panels
CLIA Waived	11 of 17 panels are CLIA waived

Parameter	Value
Measurement Method	Spectrophotometry — Xenon arc stroboscopic lamp, 9 wavelengths, 16-bit ADC
Reaction Temperature	37°C ± 1°C (maintained by internal heater during analysis)
Quality Control	Built-in iQC® (intelligent Quality Control) system — automatic
Memory	Up to 5,000 patient and QC results
Display	VGA color touchscreen
Connectivity	5 USB ports (bidirectional) + 1 Ethernet port
Data Formats	ASCII, ASTM, HL7, XML
Printer	Built-in thermal printer + external printer/LIS/EMR capable
Power Requirements	100–240 VAC, 50–60 Hz or 15 VDC, 5.0A
Dimensions	32.4 cm (H) × 14.9 cm (W) × 21.0 cm (D)
Weight	5.1 kg (analyzer) + 0.7 kg (power adapter)
Operating Temp	15–32°C, 8–80% humidity (non-condensing)
Operation Mode	Continuous, walk-away

3.3 Generated Data — 31 Chemistry Tests

The Piccolo Xpress generates up to 31 distinct chemistry results across 17 panel configurations. The following table lists all 31 analytes with their LOINC codes and clinical significance:

Analyte	Abbrev.	LOINC Code	Unit	Clinical Area
Albumin	ALB	1751-7	g/dL	Liver function, nutrition
Alkaline Phosphatase	ALP	6768-6	U/L	Liver, bone disease
Alanine Aminotransferase	ALT	1742-6	U/L	Liver damage, hepatitis
Amylase	AMY	1798-8	U/L	Pancreatitis
Aspartate Aminotransferase	AST	1920-8	U/L	Liver, heart damage
Direct Bilirubin	DBIL	1968-7	mg/dL	Liver, bile duct disease
Total Bilirubin	TBIL	1975-2	mg/dL	Liver function, jaundice
Blood Urea Nitrogen	BUN	3094-0	mg/dL	Kidney function
Total Calcium	Ca	17861-6	mg/dL	Bone, parathyroid, kidney
Creatine Kinase	CK	2157-6	U/L	Heart attack, muscle damage
Chloride	Cl ⁻	2075-0	mmol/L	Electrolyte balance
Creatinine	CRE	2160-0	mg/dL	Kidney function

Analyte	Abbrev.	LOINC Code	Unit	Clinical Area
C-Reactive Protein	CRP	1988-5	mg/L	Inflammation, infection
Gamma-Glutamyl Transferase	GGT	2324-2	U/L	Liver function, alcohol use
Glucose	GLU	2345-7	mg/dL	Diabetes, metabolic disorders
Potassium	K ⁺	2823-3	mmol/L	Cardiac, renal, metabolic
Lactate	LAC	2524-7	mmol/L	Sepsis, tissue hypoxia
Lactate Dehydrogenase	LDH	2532-0	U/L	Tissue damage, hemolysis
Magnesium	Mg	2601-3	mg/dL	Neuromuscular function
Sodium	Na ⁺	2951-2	mmol/L	Fluid balance, hyponatremia
Phosphorus	PHOS	2777-1	mg/dL	Bone, kidney, parathyroid
Total Carbon Dioxide	tCO ₂	2028-9	mmol/L	Acid-base balance
Total Protein	TP	2885-2	g/dL	Nutrition, liver, kidney
Uric Acid	UA	3084-1	mg/dL	Gout, kidney stones
Total Cholesterol	CHOL	2093-3	mg/dL	Cardiovascular risk
HDL Cholesterol	HDL	2085-9	mg/dL	Cardiovascular protection
LDL Cholesterol*	LDL*	13457-7	mg/dL	Cardiovascular risk (calc.)
Non-HDL Cholesterol*	nHDL*	70204-3	mg/dL	Cardiovascular risk (calc.)
Triglycerides	TRIG	2571-8	mg/dL	Lipid metabolism, cardiac
VLDL Cholesterol*	VLDL*	13458-5	mg/dL	Cardiovascular risk (calc.)
Cholesterol/HDL Ratio*	CHOL/HDL*	9830-1	ratio	Cardiovascular risk ratio (calc.)

* Calculated result, not directly measured.

3.4 Panel Configurations

The 31 analytes above are organized into 17 pre-configured panel configurations:

Panel Name	Catalogue #	Tests Included	CLIA Waived?
General Chemistry 6	400-0006	ALT, AST, BUN, CRE, GGT, GLU	Yes
General Chemistry 13	400-0029	ALB, ALP, ALT, AMY, AST, BUN, Ca, CRE, GGT, GLU, TBIL, TP, UA	Yes
BioChemistry Panel Plus	400-0035	ALB, ALP, ALT, AMY, AST, BUN, Ca, CRE, CRP, GGT, GLU, TP, UA	No
AmLyte 13	400-0041	ALB, ALT, AMY, AST, BUN, Ca, CK, CRE, CRP, GLU, K ⁺ , Na ⁺ , TBIL	No

Panel Name	Catalogue #	Tests Included	CLIA Waived?
Basic Metabolic Panel	400-0024	BUN, Ca, Cl-, CRE, GLU, K+, Na+, tCO2	Yes
Basic Metabolic Panel Plus	400-0031	BUN, Ca, Cl-, CRE, GLU, K+, LDH, Mg, Na+, tCO2	Yes
Comprehensive Metabolic Panel	400-0028	ALB, ALP, ALT, AST, BUN, Ca, Cl-, CRE, GLU, K+, Na+, TBIL, tCO2, TP	Yes
Electrolyte Panel	400-0022	Cl-, K+, Na+, tCO2	Yes
MetLyte 8 Panel	400-0023	BUN, CK, Cl-, CRE, GLU, K+, Na+, tCO2	Yes
MetLyte Plus CRP	400-0034	BUN, CK, Cl-, CRE, CRP, GLU, K+, Na+, tCO2	No
MetLac 12 Panel	400-0037	ALB, BUN, Ca, Cl-, CRE, GLU, K+, LAC, Mg, Na+, PHOS, tCO2	No
Lipid Panel	400-0025	CHOL, CHOL/HDL*, HDL, nHDL*, LDL*, TRIG, VLDL*	Yes
Lipid Panel Plus	400-0030	ALT, AST, GLU, CHOL, CHOL/HDL*, HDL, nHDL*, LDL*, TRIG, VLDL*	Yes
Hepatic Function Panel	400-0026	ALB, ALP, ALT, AST, DBIL, TBIL, TP	No
Liver Panel Plus	400-0003	ALB, ALP, ALT, AMY, AST, GGT, TBIL, TP	Yes
Renal Function Panel	400-0027	ALB, BUN, Ca, Cl-, CRE, GLU, K+, Na+, PHOS, tCO2	Yes
Kidney Check	400-0033	BUN, CRE	Yes

3.5 HL7 Support

The Piccolo Xpress has native HL7 support built into the analyzer hardware. It can output data directly in HL7 format via its USB and Ethernet ports without requiring additional software.

The Piccolo Xpress supports the following data transmission formats natively:

- ASCII — plain text format for simple connectivity
- ASTM E1394 — the legacy laboratory instrument standard
- HL7 — Health Level 7 v2.x messaging for LIS/EMR integration
- XML — extensible markup language for structured data exchange

Connectivity is bidirectional via 5 USB ports and 1 Ethernet port. The device can receive patient demographic orders from a LIS (downlink) and transmit results back (uplink). Data parsing and translation may be required depending on the receiving system's HL7 version and message format requirements.

The Relaymed POC connectivity platform provides automated HL7 integration for the Piccolo Xpress specifically designed for physician office laboratories — eliminating the need for a full LIS by providing direct EMR connectivity with LOINC-coded results.

3.6 FHIR Support

The Piccolo Xpress does not natively output FHIR R4 resources. However, its native HL7 and XML output capabilities make it significantly easier to implement FHIR integration compared to devices with only proprietary outputs.

FHIR integration pathway for Piccolo Xpress:

- Step 1 — Piccolo Xpress outputs results in HL7 v2.x or XML format via USB or Ethernet
- Step 2 — Results are received by a POC connectivity platform (e.g., Relaymed, Cerner CareAware, Epic Rover) or middleware
- Step 3 — The middleware transforms HL7 v2.x ORU^R01 messages into FHIR R4 DiagnosticReport and Observation resources
- Step 4 — FHIR resources are posted to the hospital FHIR server for EHR integration

When transformed to FHIR R4, each Piccolo panel result generates a DiagnosticReport resource (representing the panel) containing Observation resources for each analyte, each coded with its LOINC code, UCUM unit, reference range, and abnormal flag.

3.7 LOINC Support

The Piccolo Xpress does not internally assign LOINC codes to its output. LOINC code assignment is performed by the receiving LIS, EMR, or POC connectivity platform during result mapping. The Relaymed platform, for example, automatically maps all 31 Piccolo analytes to their LOINC codes when transmitting to an EMR.

All 31 analytes generated by the Piccolo Xpress have established LOINC codes (listed in Section 3.3). The Piccolo's XML output format facilitates accurate LOINC mapping because the analyte names are transmitted as structured text rather than proprietary codes.

3.8 Connectivity Architecture

Connection Type	Protocol	Direction	Notes
USB Port (×5)	ASCII / ASTM / HL7 / XML	Bidirectional	Connect to printer, keyboard, barcode scanner, computer, or EMR/LIS
Ethernet Port (×1)	ASCII / ASTM / HL7 / XML	Bidirectional	Network connectivity LIS/EMR
Built-in Printer	Thermal print	Output only	Adhesive-backed paper — patient results with QC data
Relaymed Platform	HL7 → EMR	Bidirectional	POC connectivity software — LOINC coded results to EMR
LIS Direct	HL7 v2.x / ASTM	Bidirectional	Compatible with all major LIS platforms
FHIR R4	Not native	—	Requires external HL7→FHIR transformation middleware

4. Comparative Standards Analysis

4.1 HL7 Support Comparison

Feature	ABX Micros 60	Piccolo Xpress
Native HL7 Output	No — requires LiteDM software	Yes — native HL7 v2.x output
HL7 Version	HL7 v2.x via LiteDM bridge	HL7 v2.x (ASCII, ASTM, HL7, XML)
Message Direction	Bidirectional via LiteDM	Bidirectional (USB + Ethernet)
Physical Interface	RS-232 serial	USB (×5) + Ethernet (×1)
EMR Compatibility	Athena, Flatiron, Epic, eClinicalWorks, CureMD	All HL7-compatible EMR/LIS systems
Setup Complexity	Requires LiteDM software installation + HL7 bridge config	Plug-and-play with HL7-capable systems
Additional Software	LiteDM v3 (optional, sold separately)	Relaymed (optional POC platform)

4.2 FHIR R4 Support Comparison

Feature	ABX Micros 60	Piccolo Xpress
Native FHIR R4 API	No	No
FHIR via Middleware	Yes — via HL7→FHIR transformation	Yes — via HL7/XML→FHIR transformation
FHIR Resource Type	DiagnosticReport + Observation (×18)	DiagnosticReport + Observation (×31 max)
Easiest FHIR Path	LiteDM → Mirth Connect → HAPI FHIR	USB/Ethernet → Relaymed → FHIR server
FHIR Observation Count	18 per blood sample	Up to 31 per panel (panel-dependent)
SMART on FHIR	Not applicable at device level	Not applicable at device level

4.3 LOINC Code Coverage

Feature	ABX Micros 60	Piccolo Xpress
Native LOINC Assignment	No — assigned by LIS/EMR	No — assigned by LIS/EMR
Total Analytes	18 CBC parameters	31 chemistry analytes
LOINC Codes Available	All 18 have established LOINC codes	All 31 have established LOINC codes

Feature	ABX Micros 60	Piccolo Xpress
OBX-3 LOINC in HL7	Via LiteDM mapping configuration	Via LIS/Relaymed mapping
Code System	LOINC (LN)	LOINC (LN)
UCUM Units	All parameters use UCUM units	All analytes use UCUM units

4.4 Overall Standards Readiness Score

The following assessment summarizes each device's readiness for modern health data interoperability standards:

Standard	ABX Micros 60 Score	Piccolo Xpress Score	Notes
HL7 v2.x	★★★★☆ (3/5)	★★★★★ (5/5)	Piccolo has native HL7; Micros 60 needs LiteDM
FHIR R4	★★☆☆☆ (2/5)	★★★★☆ (3/5)	Neither native; Piccolo easier via XML output
LOINC Coding	★★★★☆ (3/5)	★★★★☆ (4/5)	Both support LOINC via LIS; Piccolo via XML is cleaner
LIS Integration	★★★★☆ (4/5)	★★★★★ (5/5)	Piccolo's multi-format + Ethernet is more flexible
Data Richness	★★★★☆ (3/5)	★★★★★ (5/5)	Piccolo generates 31 analytes across organ systems
Ease of Setup	★★★★☆ (3/5)	★★★★☆ (4/5)	Piccolo plug-and-play; Micros 60 needs LiteDM config
Overall	★★★★☆ (3/5)	★★★★☆ (4/5)	Piccolo more interoperable out of the box

5. Relevance to Smartband V001 and FHIR Integration

Both the ABX Micros 60 and the Piccolo Xpress generate laboratory data that is directly relevant to the health monitoring goals of the Smartband V001 project. Their data output patterns serve as reference models for how clinical laboratory analyzers structure and transmit measurement data.

5.1 LOINC Code Reference for Smartband V001

Several LOINC codes used by these analyzers are also applicable to Smartband V001 sensor measurements:

Smartband Measurement V001	Relevant Analyzer Data	LOINC Code	Connection
Heart Rate (MAX30102)	Not measured by these analyzers	8867-4	Shared LOINC coding approach
SpO2 (MAX30102)	Not measured by these analyzers	2708-6	Shared LOINC coding approach
Glucose (future sensor)	Piccolo Xpress: GLU	2345-7	Direct LOINC reuse from Piccolo
CRP (inflammation marker)	Piccolo Xpress: CRP	1988-5	Direct LOINC reuse from Piccolo
Hemoglobin (derived)	ABX Micros 60: HGB	718-7	Reference for wearable Hgb estimation
Electrolytes (future)	Piccolo Xpress: Na+, K+, Cl-	Multiple	Reference for future electrolyte sensor

5.2 Key Learnings for FHIR Implementation

- Both analyzers use HL7 v2.x ORU^R01 messages to transmit laboratory results — the same message type that would carry Smartband V001 data if connected via a gateway.
- Neither analyzer generates FHIR R4 natively — confirming that middleware-based FHIR transformation (as implemented in the Smartband V001 Flask server) is the industry standard approach.
- LOINC codes are universally used for laboratory result coding but are assigned by the LIS/middleware layer, not by the instrument itself — consistent with Smartband V001's FHIR gateway approach.
- The DiagnosticReport + Observation FHIR resource pattern used by both analyzer integrations is the same pattern the Smartband V001 FHIR server implements for its sensor readings.
- The Piccolo Xpress's XML output demonstrates that structured data formats (not just ASTM) are increasingly common in POC devices — supporting the Smartband V001's JSON/FHIR approach.

6. Conclusion

The HORIBA ABX Micros 60 and Abaxis Piccolo Xpress represent two generations of clinical analyzer connectivity philosophy. The Micros 60, an older design, relies on RS-232 and requires middleware (LiteDM) for HL7 connectivity, reflecting the legacy approach to laboratory instrument integration. The Piccolo Xpress, a more modern point-of-care design, offers native HL7, ASTM, XML, and ASCII output via USB and Ethernet, reflecting the industry's shift toward more open, flexible connectivity.

Neither device natively supports FHIR R4 — which is consistent with the broader laboratory instrument market, where FHIR integration is almost universally implemented at the middleware/LIS layer rather than at the instrument itself. Both devices support LOINC coding through their LIS integration pathways, with all generated analytes having established LOINC codes.

For the Smartband V001 project, these analyzers serve as important reference points. They demonstrate that the middleware-based FHIR transformation approach implemented in the Smartband V001 Flask server is the correct and industry-standard method. The LOINC codes

for analytes like glucose, CRP, and electrolytes — generated by the Piccolo Xpress — are directly reusable for future Smartband V001 sensor expansions.